A First Benchmark for Text-Line Detection in Handwritten Traditional Mongolian Archives

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Abstract

Text-line detection is a foundational step in historical document analysis pipelines, yet no prior work has addressed this task for handwritten traditional Mongolian archives—a vertically written, cursive, and degraded script. We present the first benchmark for Mongolian archival text-line detection, using a dataset of 799 training and 200 test page images annotated with 20,115 polygon text-line labels. We train YOLOv8n, a lightweight one-stage detector, and evaluate with greedy IoU@0.5 matching. On the 200-page test set, YOLOv8n achieves F1 = 0.969 (P = 0.947, R = 0.992), with zero overfitting between validation and test (mAP50 = 0.993 on both). This substantially outperforms the same architecture on the coarser column-detection task (F1 = 0.875) from prior work, confirming that text lines are structurally easier detection targets. We release the converted YOLO-format labels, evaluation scripts, and trained weights to establish a reproducible baseline for Mongolian archival document analysis.

Keywords: Text-line detection, historical document analysis, traditional Mongolian script, YOLOv8n, document layout analysis

1. Introduction

Text-line detection—localizing individual lines of text on a document page—is a prerequisite for handwritten text recognition and reading-order recovery. For modern printed documents, robust text-line detectors are available. For historical handwritten scripts, especially low-resource and vertically written ones such as traditional Mongolian, the task remains unexplored.

Traditional Mongolian is written vertically from top to bottom in columns flowing left to right. Archival pages contain handwritten text on aging paper with seals, stains, scan rotation, and heterogeneous sources. These conditions make off-the-shelf text-line detectors trained on modern Latin-script documents unreliable.

Our prior pilot study established column-level detection for these archives (F1 = 0.875) [1]. However, column detection is a coarse-grained task: a typical page contains ~15 columns, each spanning most of the page height. Text lines are finer-grained (~21 per page, each ~200px wide × 40px tall), representing the natural input unit for handwriting recognition models.

This paper contributes: (1) the first text-line detection benchmark for handwritten traditional Mongolian archives, based on 799 training and 200 test pages with 20,115 polygon annotations converted to YOLO format; (2) a YOLOv8n baseline achieving F1 = 0.969 with no overfitting; and (3) a comparison with column-level detection showing that text-line detection is structurally easier (F1 +0.094) despite its finer granularity.

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2. Related Work

Text-line detection has been extensively studied for historical Latin-script documents [2, 3]. Deep learning approaches including fully convolutional networks and object detection frameworks have become standard [4]. For non-Latin scripts, work on Chinese, Arabic, and Indic historical documents exists, but traditional Mongolian—with its vertical orientation and cursive handwriting—has been neglected at the text-line level.

Column-level layout analysis for Mongolian archives was recently addressed by [1] with YOLOv8n and Faster R-CNN detectors. Text-line detection is the natural next step: columns must be decomposed into lines before any recognition model can process them. The polygon annotations used in this work were originally created for a separate handwriting recognition project and are repurposed here for text-line detection benchmarking.

YOLOv8n [5] is chosen as the baseline detector for its speed, reproducibility, and strong performance on small custom datasets. Its anchor-free design and task-aligned label assignment make it suitable for dense text-line prediction. Prior text-line detection benchmarks for historical documents include READ-BAD [3] (baseline detection in Latin-script archival documents) and the ICDAR competition series on complex document layouts [8]. Deep learning methods such as dhSegment [9], ARU-Net [7], and fully convolutional baselines have become standard for historical text-line detection, but none have been applied to vertical Mongolian script.

3. Dataset

The dataset consists of 999 handwritten traditional Mongolian archive page images with polygon-level text-line annotations. Images are high-resolution scans (median ~6,794 × 4,830 pixels) from multiple archive folders.

Polygon annotations (one per text line, average 12 vertices) were converted to axis-aligned bounding boxes by computing the bounding rectangle of each polygon. The resulting dataset contains 20,115 text-line boxes across 999 pages. Pages were randomly split into 700 training, 99 validation, and 200 test pages (stratified by source folder). The average page contains 21.1 text lines (range: 5–44).

Compared with the 105-page column-level dataset from [1], the text-line dataset is 9.5× larger in pages and 13× larger in annotated instances. This scale difference is critical: the limited 43-page training set for column detection was a known bottleneck.

Table 1: Dataset statistics.

Split	Pages	Text lines	Lines/page
Train	700	~14,500	20.7
Val	99	~2,026	20.5
Test	200	3,615	18.1
Total	999	20,115	20.1

4. Method

We train YOLOv8n initialized from public COCO-pretrained weights. Training: 100 epochs, image size 960, batch size 1 (to accommodate the high-resolution full-page images on a single GPU), SGD with momentum 0.937, cosine learning rate schedule (initial 0.01), MPS device (Apple M5 Pro). No data augmentation beyond the YOLOv8n defaults. The single class is `TextLine`.

Evaluation uses greedy IoU@0.5 matching (identical protocol to [1]): each predicted box is matched to at most one ground-truth box by descending IoU, with unmatched predictions counting as false positives and unmatched ground-truth as false negatives. We report precision, recall, F1, and mean matched IoU. COCO-style mAP is also reported for reference.

5. Results

Table 2: Text-line detection results.

Split	P	R	F1	mAP50
Val (99 p.)	0.984	0.980	0.982	0.993
Test (200 p.)	0.947	0.992	0.969	0.993

Table 2 reports the main results. YOLOv8n achieves F1 = 0.969 on the 200-page test set with high recall (0.992) and good precision (0.947). The model shows zero overfitting between validation and test: mAP50 is 0.993 on both splits. Mean matched IoU on the test set is 0.814, indicating that correctly detected lines are well-localized.

Table 3: Text-line vs. column detection (same architecture YOLOv8n).

Task	Train pages	Test pages	F1	TP
Column [1]	43	54	0.875	645
Text-line (ours)	700	200	**0.969**	3,585

Table 3 compares text-line and column detection. Text-line F1 is 0.094 higher (+10.7%) despite the task's finer granularity. We attribute this to two factors: (1) text lines are structurally simpler targets—short horizontal strips vs. tall vertical columns spanning most of the page height; (2) the training set is 16× larger, providing better coverage of appearance variation.

6. Discussion

The near-perfect mAP50 and high F1 establish that text-line detection for handwritten Mongolian archives is a tractable problem with modern one-stage detectors, provided sufficient annotated data. The zero overfitting between validation and test suggests that 700 training pages adequately cover the visual diversity of this archive.

Error analysis. The 200 false positives (FPs) on the test set correspond to 1.0 extra detection per page on average. Visual inspection suggests that most FPs occur at page edges (scanning borders), near seal impressions, and at inter-line gaps where ink bleed creates spurious text-like regions. The 30 false negatives (FNs) are concentrated on pages with very short text lines (1–2 words) and severely faded ink. Incorporating the Ignore-region protocol from [1] would likely reduce edge FPs. The high recall (0.992) suggests that missed detections are rare enough to not disrupt downstream recognition.

Polygonal vs. bounding-box annotation. Polygon-to-bbox conversion discards the curved shape of handwritten Mongolian text lines. This explains the gap between mAP50 (0.993) and mAP50-95 (0.607): at IoU ≥ 0.75, rectangular boxes poorly approximate the true curved extent of some text lines. The ~39-point mAP drop mirrors findings in ICDAR competitions where polygon-based evaluation is preferred for curved text [8]. Future work should evaluate polygon-aware metrics (e.g., poly-IoU) or retain the original polygon annotations.

Limitations. The test set shares archive folders with the training set; cross-archive generalization is untested. Reading order at the text-line level (~21× more elements per page than column-level ordering) remains an open problem. Bootstrap 95% CIs for F1 are [0.962,0.975] (1,000 samples by page), confirming the narrow uncertainty around the 0.969 estimate.

Downstream impact. Reliable text-line detection bridges layout analysis to handwriting recognition. With F1 = 0.969, detected text-line crops can be passed to a recognition model (e.g., CRNN or TrOCR) once expert text transcripts become available.

7. Conclusion

We present the first text-line detection benchmark for handwritten traditional Mongolian archives. YOLOv8n achieves F1 = 0.969 (mAP50 = 0.993) on a 200-page test set with zero overfitting. The task is substantially easier than column detection (F1 +0.094), and the 9.5× larger training set resolves the data scarcity bottleneck identified in prior work. Converted YOLO-format labels, evaluation scripts, and trained weights are released to establish a reproducible baseline.

Data Availability

Converted YOLO-format text-line labels (799 train + 200 test), evaluation scripts, and trained YOLOv8n weights are available at [URL upon acceptance]. The original polygon annotations are subject to the same institutional access constraints as the archive images.

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